

# ELECTRON DIFFRACTION

## INTRODUCTION



This experiment is a demonstration of the wave nature of the electron, and provides a confirmation of the de Broglie relationship:

$$\lambda = \frac{h}{p} \quad (1)$$

where  $\lambda$  = electron wavelength,  $h$  = Planck's constant,  $p$  = electron momentum. It also provides an introduction to the use of diffraction in the analysis of crystals.

This guide sheet outlines a method for the analysis of cubic crystal forms, this being useful to you for interpreting the transmission diffraction pattern produced by scattering electrons off a thin film target of polycrystalline aluminium. The apparatus also contains samples with hexagonal structures. These are pyrolytic graphite targets, and are available both as single crystals and in polycrystalline form.

For the methodology of analysis of the hexagonal crystal, and for additional material on cubic crystals, see the reference.

### THE DE BROGLIE WAVELENGTH

The voltmeter measures the accelerating potential of the electrons in the tube. Thus:

$$\frac{1}{2} m v_f^2 = eV \quad \text{or} \quad p = m v_f = \sqrt{2meV} \quad (2)$$

where  $v_f$  is the final velocity of the electrons after being accelerated through a potential  $V$ . The above assumes the non-relativistic approximation. To what degree is this justified?

Substituting in the de Broglie relationship, equation(1);

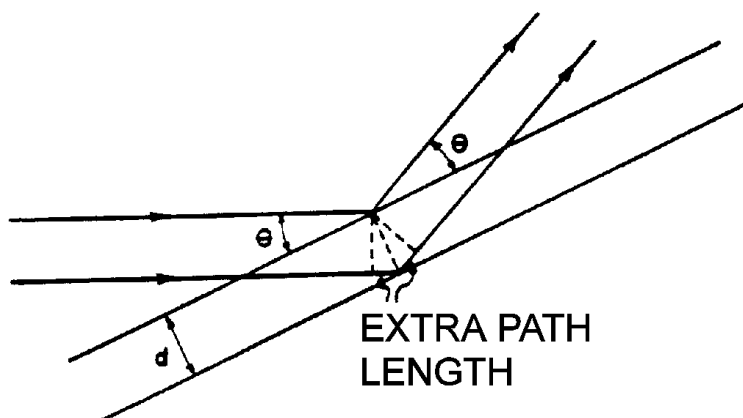
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}} = \sqrt{\frac{h^2/2me}{V}} \quad (3)$$

When the values of  $h$ ,  $m$ , and  $e$  are substituted:

$$\lambda \text{ (nm)} = \sqrt{\frac{1.505}{V(\text{Volts})}} \quad (4)$$

## BRAGG'S LAW

The case of waves (electromagnetic waves such as x-rays or "matter" waves such as electrons) scattering off a crystal lattice is similar to light being scattered by a diffraction grating. However, the three-dimensional case of the crystal is geometrically more complex than the two- (or one-) dimensional diffraction grating case. Bragg's Law governs the position of the diffracted maxima in the case of the crystal. A wave diffracted by a crystal behaves as if it were reflected off the planes of the crystal. Moreover there is an outgoing diffracted wave only if the path length difference between rays "reflected" off adjacent planes are an integral number of wavelengths. Thus considering a beam scattering off two parallel planes of atoms as shown in figure.



A beam incident on a pair of planes separated by a distance  $d$ . (For reinforcement of the scattering from atoms in **one** plane the usual condition for reflection applies, - angle of reflection equal angle of incidence, as indicated.)

The extra path length of the lower ray may be shown to be  $2d \sin \theta$  so that maxima in the diffraction pattern will occur when:

$$2d \sin \theta = n\lambda, \quad n = 0, 1, 2, \dots \quad (5)$$

This is Bragg's Law. Furthermore, the beam is deflected a total angle  $2\theta$ . Thus, for our electron diffraction tube, with maxima registered as spots or rings on the face of the tube, the distance of the spot from the incoming beam axis =  $R$ , so

$$R = D \tan(\text{deflection}) = D \tan 2\theta \sim D \cdot 2\theta \quad (6)$$

where  $D$  = distance from target to screen. Combining equations (5) and (6), and taking  $\sin \theta \approx \theta$ , then:

$$R = \frac{n\lambda D}{d} \quad (7)$$

(Note that for the polycrystalline samples mentioned below,  $r$  is the radius of the ring.)

Note then, that the obtaining of a diffraction maximum requires that two conditions be met. Not only must the angle of deflection bear an appropriate relationship to  $d$  and  $\lambda$ , but also the crystal orientation must be correct to provide an apparent "reflection" off the crystal planes. The way the crystals are oriented relative to the incoming beam will thus determine the appearance of the diffraction pattern,

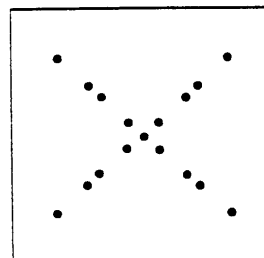
## ELECTRON DIFFRACTION PATTERNS

In relation to diffraction patterns it is interesting to consider three types of solid matter: single crystals, polycrystals and amorphous materials.

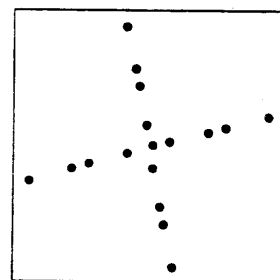
### SINGLE CRYSTALS

Single crystals consist of atoms arranged in an orderly lattice. Some types of crystal lattices are simple cubic, face centre cubic (f.c.c.), and body centre cubic (b.c.c). In general, single crystals with different crystal structures will cleave into their own characteristic geometry. You may have seen single crystals of quartz, calcite, or carbon (diamond).

Single crystals are the most ordered of the three structures. An electron beam passing through a single crystal will produce a pattern of spots. From the diffraction spots one can determine the type of crystal structure (f.c.c., b.c.c.) and the "lattice parameter" (*i.e.*, the distance between adjacent (100) planes).



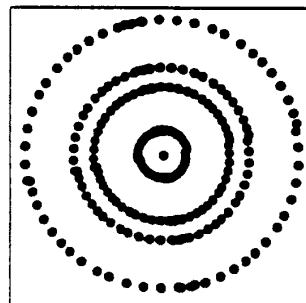
Also, the orientation of the single crystal can be determined: if the single crystal is turned or flipped, the spot diffraction pattern will rotate around the centre beam spot in a predictable way.



### POLYCRYSTALLINE MATERIALS

Polycrystalline materials are made up of many tiny single crystals. Most common metal materials (copper pipes, nickel coins, stainless steel forks) are polycrystalline. Also, a ground-up powder sample appears polycrystalline. Any small single crystal "grain" will not in general have the same orientation as its neighbours. The single crystal grains in a polycrystal will have a random distribution of all the possible orientations.

A polycrystal, therefore, is not as ordered as a single crystal. An electron beam passing through a polycrystal will produce a diffraction pattern equivalent to that produced by a beam passing through series of single crystals of various orientations. The diffraction pattern will therefore look like a superposition of single crystal spot patterns: a series of concentric rings resulting from many spots very close together at various rotations around the centre beam spot. From the diffraction rings one can also determine the type of crystal structure and the "lattice parameter". One cannot determine the orientation of a polycrystal, since there is no single orientation and flipping or turning the polycrystal will yield the same ring pattern.

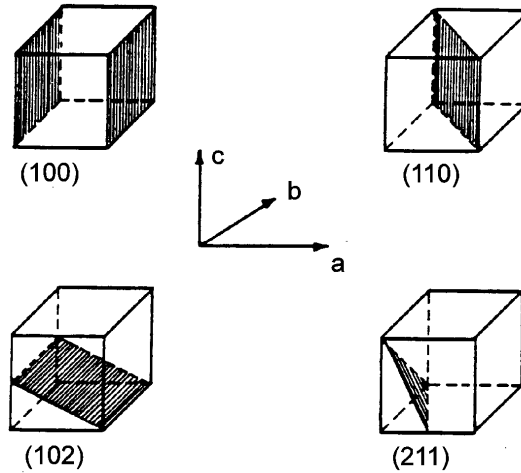


## AMORPHOUS MATERIALS

Amorphous materials do not consist of atoms arranged in ordered lattices, but in hodgepodge random sites. Amorphous materials are completely disordered. The electron diffraction pattern will consist of fuzzy rings of light on the fluorescent screen. The diameters of these rings of light are related to average nearest neighbour distances in the material.

## THE MILLER INDICES FOR CUBIC CRYSTALS

The Miller indices characterize various planes through a crystal lattice. First choose crystallographic axes,  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  with the origin at one atom. The Miller indices are defined to be the reciprocals of the fractional intercept of the plane with the three axes, as shown in the figure. If the plane is parallel to a given axis, the index is = 0, corresponding to an intercept of infinity.



Miller indices of some lattice planes

For a cubic crystal, the distance of the specified plane from the parallel plane which passes through the origin is:

$$d = \frac{a}{\sqrt{H^2 + K^2 + L^2}} \quad (8)$$

where  $H, K, L$  are the Miller indices and  $a$  = length of a side of the cube. The proof of this is a simple matter of solid geometry.

The lower the Miller indices, the greater the separation between adjacent parallel planes. In addition, the lower indexed planes have a greater density of lattice points.

The aluminium target is a powder of small crystal flakes, *i.e.*, it is polycrystalline. Thus the diffraction pattern appears as rings. There are also many possible planes that may be drawn through the crystal lattice, and in a powder sample in general all of these will be present. A simple example of this is shown in Figure 1.

A more useful form (as shown below) may be found by multiplying numerator and denominator by the order of interference,  $n$ :

$$d = \frac{na}{\sqrt{h^2 + k^2 + \ell^2}} \quad (9)$$

where  $h = nH, k = nK$  and  $\ell = nL$ ;  $n, k$ , and  $\ell$  are also integers.

## EXPERIMENT I. THE ELECTRON DIFFRACTION PATTERN OF POLYCRYSTALLINE ALUMINUM PROCEDURE

1. Once the electron diffraction tube has warmed up focus the beam and aim it at the piece of the polycrystalline aluminum foil. When the beam is pointed on the foil a bright set of rings should appear.
2. For three different anode voltage levels (suggested values are 8.5, 9, and 9.5 Kilovolts) find the radii of the rings produced. To obtain good data, take two or three measurements of the diameter of each ring using different orientations and average them. This will help compensate for distortions of the screen, variations in intensities, and limited accuracy of our eyes.
3. When you vary the anode voltage it will help to change it in small increments while readjusting the beam position and focus to maintain the ring pattern until the desired voltage is reached. Don't forget to monitor and adjust the anode voltage while making your measurements since it may float a bit with time.
4. **The distance D between the crystal and the screen is 18.29 cm.**
5. Using this data you will perform two sets of analysis:

### CALCULATIONS

Calculation of the beam wavelength For each accelerating (anode) voltage perform the following:

1. Calculate the wavelength of the electron beam from the accelerating voltage using Equation

$$\lambda \text{ (Å)} = \sqrt{\frac{150}{V \text{ (Volts)}}}$$

2. Using the **lattice constant of aluminum (a = 4.0496 Å)** and the measured radii of the rings, match each ring to the correct reflection plane and its corresponding Miller indices using the Equation:

$$\lambda = \frac{2 a \sin \theta}{\sqrt{h^2 + k^2 + l^2}} = \frac{a r}{D \sqrt{h^2 + k^2 + l^2}}.$$

The allowed lattice spacing given in Table 1 will be necessary to identify each ring. It may be a little tricky. Keep in mind that it is possible a ring may be missing or very dim. Tabulate all values.

Table 1 Allowed Face-Center Cubic Reflections

| $h \ k \ l$  | $(h^2 + k^2 + l^2)$ | $\sqrt{h^2 + k^2 + l^2}$ |
|--------------|---------------------|--------------------------|
| 1 1 1        | 3                   | 1.732                    |
| 2 0 0        | 4                   | 2.000                    |
| 2 2 0        | 8                   | 2.828                    |
| 3 1 1        | 11                  | 3.316                    |
| 4 0 0        | 16                  | 4.000                    |
| 3 3 1        | 19                  | 4.358                    |
| 4 2 0        | 20                  | 4.472                    |
| 4 2 2        | 24                  | 4.898                    |
| 5 1 1, 3 3 3 | 27                  | 5.198                    |
| 4 4 0        | 32                  | 5.656                    |

3. Once each ring has been identified, using the accepted value for the lattice constant, recalculate the wavelength of the electron beam based on the measured ring radii. Calculate and tabulate the percent error and discuss.

## EXPERIMENT II. THE ELECTRON DIFFRACTION PATTERN OF HEXAGONAL PYROLYTIC GRAPHITE

### PROCEDURE

Aim the electron beam on to the piece of graphite to produce a hexagonal pattern of spots. You may need to play the beam around the edge and while adjusting the focus to find a thin area of the crystal which produces a sharp pattern.

For three different anode voltage levels (suggested values are 8.5, 9, and 9.5 Kilovolts) record the distance from the inner spot to nearest six spots. If it is easier, measure the distance between opposing spots of the pattern. For each anode voltage average these values and use the relation

$$\sin \theta \approx \theta \approx \frac{r}{2D}$$

to obtain the first order diffraction angle at the electron beam wavelength corresponding to that particular voltage. This gives the diffraction angle corresponding to the largest fundamental lattice plane spacing of the crystal.

When you vary the anode voltage it will help to change it in small increments while readjusting the beam position and focus to maintain the spot pattern until the desired voltage is reached. Don't forget to monitor and adjust the anode voltage while making your measurements since it may float a bit with time.

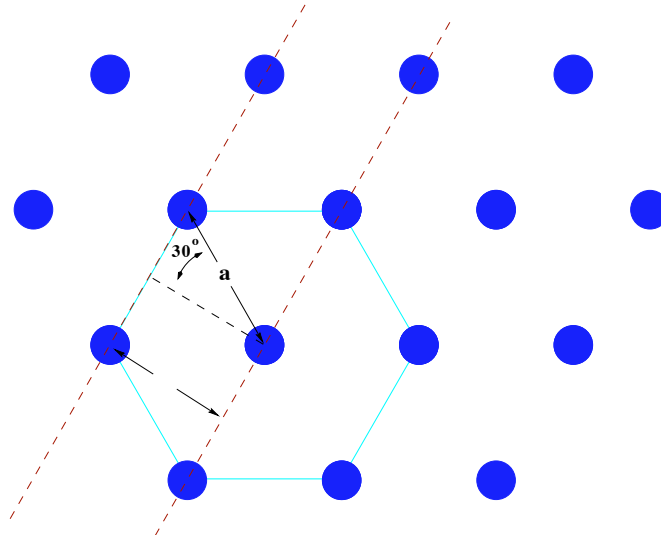


Figure: Lattice plane spacing of the Fundamental Lattice Planes with the Widest Spacing for Hexagonal Crystal Structures

## CALCULATIONS

### A. CALCULATION OF THE LATTICE CONSTANT

For each accelerating (anode) voltage perform the following,

1) The set of fundamental lattice planes with the widest separations is shown in Figure above. For these lattice planes the spacing  $d$  is related to the lattice constant  $a$  by,

$$d = a \cos(30^\circ)$$

For each accelerating voltage calculate the lattice spacing using the calculated values of the diffraction angles

$$\left( \theta \approx \frac{r}{2D} \right)$$

The Bragg's equation (with  $n = 1$ ), and the beam wavelength calculated from the accelerating

potential  $\left( \lambda \text{ (Å)} = \sqrt{\frac{150}{\text{Volts}}} \right)$

2) Compare the calculated values for the lattice spacing with the accepted value  $a = 2.4612 \pm 0.0001 \text{ Å}$ .

3) By substituting deBroglie's relation into the Bragg Equation we can use the data from this experiment to estimate Planck's constant. For first order diffraction ( $n=1$ ) and small  $\theta$

$$\lambda = \frac{h}{p} = 2 d \sin \theta \approx 2 d \frac{r}{D} = \frac{d r}{D}.$$

after substituting the equation

$$\text{K.E.} = \frac{1}{2} m v^2 = \frac{p^2}{2m} = q V$$

and solving for "r" produces

$$r = h \frac{D}{d \sqrt{2 q m V}}$$

Planck's constant may be found from the slope of a line fit to a plot of r as a function of

$$\frac{D}{d \sqrt{2 q m V}}$$